

September Plan: Quality Measures × Inferno US Quality Core Elements

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Audience: engineering + Connectathon prep

Related: STRATEGY.md , CMS_2026_QUALITY_MEASURES.md ,
SEPTEMBER_CONNECTATHON_RECOMMENDATIONS.md , QUALITY_AI_CONSULT_PLAN.md , Vista-on-FHIR docs/
US_QUALITY_CORE_INFERNO.md

Goal

Pass as much of the hosted **Inferno US Quality Core Server v0.5.0** suite as possible by September, using **legitimate CMS eCQM test patients** (denominator + numerator), not synthetic filler-only charts.

Working idea:

Turn on a small set of quality measures and keep **one strong patient per measure**. Taken together, those patients should exercise the Inferno profile groups that are still missing or skipped today.

This document maps each first-wave measure to the Inferno data-element / profile groups it naturally requires, identifies **new** elements beyond what CMS165 + CMS122 already cover, and recommends which measures to prioritize for September.

Two Success Criteria (Do Not Collapse Them)

Layer	Question	Tool
Clinical legitimacy	Is the patient in IPP / DENOM / NUMER for a real 2026 eCQM?	CQL / DEQM MeasureReport
FHIR Quality API	Does /fhir (and /altfhir source) satisfy US Quality Core search, read, profile, Must Support, and Provenance?	Inferno us_quality_core_v050

A September demo patient must clear **both**. Green Inferno with a clinically nonsense chart is not the goal; green CQL with no Inferno coverage is also not enough.

Inferno Surface (What “Data Elements” Means Here)

Inferno US Quality Core v0.5.0 currently generates **58 profile groups**. For planning we treat each group as a **data-element family**: resource type + profile semantics that Inferno will search, read, validate, and often Provenance-`_revinclude`.

Families already exercised well by CMS165 / CMS122 work (and VistA vitals/labs):

Inferno group	Why it already shows up
Patient	Every cohort
Encounter	Qualifying visits
Condition Problems Health Concerns	HTN / diabetes problem list
Condition Encounter Diagnosis	Visit POV / encounter diagnosis
Observation US Core Blood Pressure	CMS165 numerator
Observation Laboratory Result	CMS122 HbA1c
DiagnosticReport for Laboratory Results Reporting	Lab panels / HbA1c report
DocumentReference	Progress / Quality AI Consult notes
Organization / Practitioner / Location	Reference resolution (partial)
Provenance	Cross-cutting; still incomplete

Families that still drive large skip counts on showcase patients (see `QUALITY_AI_CONSULT_PLAN.md` skip analysis) and need **measure-motivated** data:

Inferno group	Typical measure driver
Observation US Core Smoking Status	CMS138
Observation Screening Assessment	CMS2 (PHQ / depression screen)
Observation Clinical Result	CMS131 eye exam / clinical findings
Procedure / Procedure Not Done	CMS130, CMS125, CMS131, CMS138 counseling
ServiceRequest / Service Not Requested	Follow-up orders (CMS2, CMS22, CMS125, CMS130)
CarePlan / Goal / Task	Follow-up plan after positive screen (CMS2)
MedicationRequest / Medication Not Requested	CMS68, CMS138 cessation meds
Immunization / Immunization Not Done	Immunization-family / CMS22 substitute slot
DiagnosticReport for Report and Note Exchange	Mammography / clinical notes as DR Note
AllergyIntolerance	Often present in Synthea; keep on at least one patient
Specimen	Lab pathway completeness (HbA1c / FIT)

Out of scope for September measure-driven coverage unless leftover capacity (not needed by the first-wave eQMs):

- AdverseEvent, Coverage, DeviceRequest(+Not), FamilyMemberHistory, RelatedPerson
- Pregnancy Intent/Status, Occupation, pediatric BMI/weight/OFC
- MedicationAdministration / MedicationDispense (+ declined/not-done variants)
- Observation Cancelled, Simple Observation (unless a measure forces it)

Those can still be filled later with synthetic resources; they should not choose the September measure set.

Measure → Inferno Element Mapping

For each measure: clinical facts required for legitimate DENOM/NUMER, Inferno groups those facts exercise, and **new** groups relative to a CMS165+CMS122 baseline.

Legend for **New vs baseline**:

- **Baseline** = already targeted by CMS165/CMS122 + current VistA vitals/labs
- **New** = primary reason to turn this measure on for Inferno coverage
- **Shared** = needed by the measure but already in baseline (still must be present on that patient)

Mapping table (measures × Inferno groups)

Inferno group (US Quality Core v0.5.0)	CMS165 BP	CMS122 A1c	CMS130 CRC	CMS125 Breast	CMS138 Tobacco	CMS2 Depression	CMS131 Eye	CMS127 M
Patient	Shared	Shared	Shared	Shared	Shared	Shared	Shared	SH
Encounter	Shared	Shared	Shared	Shared	Shared	Shared	Shared	SH
Condition Problems / Encounter Dx	HTN	Diabetes	Shared‡	Shared‡	Shared‡	Shared‡	Diabetes	SH
Observation Blood Pressure	Baseline	—	—	—	—	—	—	—
Observation Laboratory Result	—	Baseline (HbA1c)	New (FIT/FOBT/stool DNA)	—	—	—	—	—
DiagnosticReport Lab	—	Baseline	New (FIT panel)	—	—	—	—	—
Specimen	—	Shared	Shared	—	—	—	—	—
Observation Smoking Status	—	—	—	—	New	—	—	—
Observation Screening Assessment	—	—	—	—	—	New (PHQ)	—	—
Observation Clinical Result	—	—	—	—	—	—	New (retinal)	—
Procedure	—	—	New (colonoscopy / flex sig)	New (mammography)	New (cessation counseling)	Shared (follow-up)	New (eye exam)	SH re
Procedure Not Done	—	—	Optional	Optional	Optional	Optional	Optional	—
ServiceRequest / Not Requested	—	—	Optional order	New (imaging order)	Optional	New (referral)	Optional	—

Inferno group (US Quality Core v0.5.0)	CMS165 BP	CMS122 A1c	CMS130 CRC	CMS125 Breast	CMS138 Tobacco	CMS2 Depression	CMS131 Eye	CMS147 Immunization
CarePlan / Goal / Task	—	—	—	—	—	New (follow-up plan)	—	—
MedicationRequest / Not Requested	—	—	—	—	New (cessation Rx)	—	—	New (d m)
DiagnosticReport Note	—	—	—	New (mammo report)	—	—	Shared	—
DocumentReference	Shared	Shared	Shared	Shared	Shared	Shared	Shared	New lis
Immunization / Not Done	—	—	—	—	—	—	—	—
AllergyIntolerance	Keep on ≥1 patient (Synthea)							
Provenance	Infrastructure for all patients / all groups							

† CMS22v14 is the 2026 substitute for the withdrawn CMS147 immunization slot in the EC QDM ZIP (CMS_2026_QUALITY_MEASURES.md).

‡ Qualifying encounter / age-sex population; problem list not always measure-critical but helps Inferno Condition groups.

†† Prefer a dedicated immunization patient if CMS22 alone does not force Immunization resources.

Per-measure “new elements” summary

Measure	Clinical DENOM / NUMER gist	New Inferno families unlocked	Loader / helper risk
CMS165v14 Controlling High Blood Pressure	HTN + ambulatory encounter; controlled BP	(baseline) BP Observation components	Low — vitals path exists (C0FWVIT)
CMS122v14 Diabetes Glycemic Status >9%	Diabetes + encounter; HbA1c result	(baseline) Lab Observation; Lab DiagnosticReport	Medium — C0FWLAB / AFICN fixed; keep Specimen/DR tight
CMS130v14 Colorectal Cancer Screening	Age 45–75 + encounter; colonoscopy / FIT / stool DNA	Procedure; FIT Lab Obs/DR	High — Procedure + lab stool tests must survive load

Measure	Clinical DENOM / NUMER gist	New Inferno families unlocked	Loader / helper risk
CMS125v14 Breast Cancer Screening	Female age band + encounter; mammography	Procedure; DiagnosticReport Note; ServiceRequest	High — imaging/note DR historically thin in VistA FHIR
CMS138v14 Tobacco Screening & Cessation	Encounter; smoking status; intervention if user	Smoking Status ; counseling Procedure; cessation MedicationRequest	Medium — social-history Obs often missing on / fhir
CMS2v15 Depression Screening & Follow-Up	Age ≥12 + encounter; screen; follow-up if positive	Screening Assessment ; CarePlan/ServiceRequest	High — new Observation category + follow-up artifacts
CMS131v14 Diabetes Eye Exam	Diabetes + encounter; retinal/eye exam	Procedure; Clinical Result Observation	Medium — clinical-result profile distinct from lab
CMS68v15 Documentation of Current Medications	Encounter; med list documented	MedicationRequest completeness; DocumentReference	Medium — MedRequest search/profile gaps
CMS22v14 High BP Screening & Follow-Up	Encounter; BP screen; follow-up when indicated	ServiceRequest / follow-up Procedure (overlaps CMS165)	Medium — follow-up documentation is the new part

Coverage Plan: One Patient Per Measure

Selection rule

For each activated measure, freeze **one showcase patient** that is:

1. In DENOM and NUMER on official (or workspace-verified) CQL for the 2026 definition.
2. Loaded to VistA (load=1) with numerator evidence still visible on /fhir .
3. Hosted-Inferno tested as part of a **multi-patient** Inferno session (comma-separated Patient IDs), so skipped groups on patient A can pass on patient B.

Do **not** expect one patient to green the whole suite. Inferno skips when the selected patients collectively lack a profile family.

Recommended September portfolio (priority order)

Priority	Measure	Showcase role	Primary new Inferno coverage	Engineering status (2026-07-23)
1	CMS165v14	Anchor vitals / HTN	Blood Pressure (+ Condition/ Encounter)	Active; CQL path proven on selected-18; Quality AI Consult BP writeback
2	CMS122v14	Anchor labs / diabetes	Lab Observation + Lab DiagnosticReport	Active; Quality AI Consult HbA1c via C0FWLAB
3	CMS138v14	Unlock smoking status	Smoking Status (+ cessation Procedure/MedRequest)	Cohort pending; high Inferno ROI

Priority	Measure	Showcase role	Primary new Inferno coverage	Engineering status (2026-07-23)
4	CMS2v15	Unlock screening Obs + follow-up	Screening Assessment + CarePlan/ServiceRequest	Cohort pending; high Inferno ROI
5	CMS130v14	Unlock screening Procedure + FIT	Procedure + stool Lab/DR	Cohort pending; high clinical visibility
6	CMS125v14	Unlock note/imaging DR	DiagnosticReport Note + mammography Procedure	Cohort pending; harder VistA round-trip
7	CMS131v14	Unlock clinical-result Obs	Observation Clinical Result + eye Procedure	After 130/125 Procedure path exists
8	CMS68v15	Harden MedicationRequest	MedicationRequest Must Support / code search	After med cache/profile work
9	CMS22v14	Follow-up orders (if imm slot empty)	ServiceRequest follow-up; optional Immunization patient	Only if Immunization still uncovered

September “turn on” target: priorities **1-6** as active dashboard measures, each with one Inferno-ready DFN. Priorities 7-9 are stretch.

Collective Inferno coverage (what the set buys)

With one good patient each for CMS165, CMS122, CMS138, CMS2, CMS130, CMS125:

Inferno family	Covered by
Patient, Encounter, Condition	All
Blood Pressure	CMS165 (+ CMS22 if used)
Lab Observation / Lab DR / Specimen	CMS122 + CMS130 (FIT)
Smoking Status	CMS138
Screening Assessment	CMS2
Procedure	CMS130, CMS125, CMS138, CMS2
ServiceRequest	CMS2, CMS125
CarePlan / Goal / Task	CMS2
MedicationRequest	CMS138, later CMS68
DiagnosticReport Note	CMS125
DocumentReference	All (notes / Quality AI Consult)
Provenance	Infrastructure on all

Still likely thin after that set (accept as September backlog unless synthetic):

- Immunization / Not Done (add a dedicated imm patient if needed)
- Observation Clinical Result (CMS131 stretch)
- AdverseEvent, Coverage, Device*, FamilyMemberHistory, pediatric/pregnancy/ occupation groups

Engineering Implications

Helpers / loader work ordered by measure ROI

Work item	Unlocks measures	Inferno groups
Keep BP vitals + HTN Condition solid	CMS165, CMS22	Blood Pressure, Condition, Encounter
Keep HbA1c lab + AFICN/ICN path solid	CMS122	Lab Observation, Lab DR, Specimen
Smoking-status Observation on /fhir	CMS138	Smoking Status
Screening-assessment Observation (PHQ LOINC)	CMS2	Observation Screening Assessment
Procedure load/read for colonoscopy, mammo, eye, counseling	CMS130, CMS125, CMS131, CMS138	Procedure
ServiceRequest (referral / imaging / follow-up)	CMS2, CMS125, CMS22, CMS130	ServiceRequest
CarePlan or Task for positive depression follow-up	CMS2	CarePlan / Task
DiagnosticReport Note for mammography	CMS125	DiagnosticReport Note
MedicationRequest profile + code search	CMS68, CMS138	MedicationRequest
Provenance generation + cache invalidation	All	Provenance _revinclude

Validation loop (per measure patient)

1. CQL MeasureReport: IPP/DENOM/NUMER.
2. Hosted Inferno on /altfhir (source).
3. load=1 into Vista.
4. Hosted Inferno on /fhir (round-trip).
5. Record pass/fail/skip delta; only promote to September demo set when clinical + /fhir Inferno are both acceptable.

Track results in SEPTEMBER_CONNECTATHON_RECOMMENDATIONS.md .

Suggested Inferno Multi-Patient Session Shape

Once showcase DFNs exist:

```
patient_ids = <CMS165>,<CMS122>,<CMS138>,<CMS2>,<CMS130>,<CMS125>
```

Run full us_quality_core_v050 against <https://devfhir.vistaplex.org/fhir> (and the /altfhir source set for the same graph IENs). Compare skip families before/after each new measure patient is added — the skip count for that family should drop when the new patient contributes the missing resource type.

Decision Summary

1. **Keep CMS165 + CMS122 as the baseline pair** (vitals + labs + conditions).
2. **Add CMS138 and CMS2 next** — they unlock unique Inferno Observation profiles (Smoking Status, Screening Assessment) and follow-up artifacts.
3. **Add CMS130 and CMS125** to force Procedure + FIT lab + DiagnosticReport Note pathways.
4. **Treat CMS131, CMS68, CMS22/immunization as stretch** once Procedure/MedRequest/ServiceRequest helpers exist.
5. Use **one legitimate NUMER patient per activated measure**, combined in one Inferno session, to cover missing elements collectively rather than overloading a single chart.

Open Items Before Freeze

- Confirm 2026 kidney-health replacement if that slot is still desired (CMS134 absent from EC QDM ZIP).
- Decide whether Immunization coverage is a seventh showcase patient or folded into CMS22.
- Produce the six showcase DFN/IEN pairs and paste them into `SEPTEMBER_CONNECTATHON_RECOMMENDATIONS.md`.
- Re-run hosted Inferno after each patient is added; update the mapping table's "engineering status" column.

Appendix A — Full Inferno Group Catalog (v0.5.0)

Generated from `us-quality-core-test-kit metadata.yml` (58 groups):

AdverseEvent; AllergyIntolerance; CarePlan; CareTeam; Condition Encounter Diagnosis; Condition Problems Health Concerns; Coverage; DeviceRequest; Device Not Requested; DiagnosticReport Note; DiagnosticReport Lab; DocumentReference; Encounter; FamilyMemberHistory; Goal; Immunization; Immunization Not Done; Location; Medication; MedicationAdministration; MedicationAdministration Not Done; MedicationDispense; MedicationDispense Declined; MedicationRequest; Medication Not Requested; Observation Clinical Result; Observation Simple; Observation Screening Assessment; Observation Cancelled; Pregnancy Intent; Pregnancy Status; Smoking Status; Observation Lab; Occupation; Blood Pressure; BMI; Pediatric BMI for Age; Body Height; Body Temperature; Body Weight; Head OFC Percentile; Heart Rate; Pulse Oximetry; Respiratory Rate; Pediatric Weight for Height; Organization; Patient; Practitioner; PractitionerRole; Procedure; Procedure Not Done; Provenance; RelatedPerson; ServiceRequest; Service Not Requested; Specimen; Task; Task Rejected.

Appendix B — First-Wave Measure IDs

From `CMS_2026_QUALITY_MEASURES.md` / `2026/measures/`:

CMS165v14 , CMS122v14 , CMS130v14 , CMS125v14 , CMS138v14 , CMS2v15 , CMS131v14 , CMS68v15 , CMS22v14 (plus reserved slots CMS134/CMS147 marked absent or substituted in the 2026 EC ZIP).